

Technical Report

EuMINE DataBridge Hackathon 2026
Team: **CataLIST**

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June 2026

1 Introduction

We present a multi-model ensemble for predicting formation energy per atom and electronic band gap of inorganic crystalline materials from crystal structure alone. The core architecture combines two complementary graph neural networks — ALIGNN [1] and MACE-MP-0 [2] — with an Optuna-optimized weighted ensemble and a post-hoc isotonic calibration layer. Both models are pretrained on large-scale DFT databases and fine-tuned on the 850-structure EuMINE Bridge Dataset.

The design philosophy prioritizes: (1) complementary inductive biases (bond-angle vs. equivariant), (2) honest uncertainty quantification via model disagreement, and (3) full MatFed API v1 compliance for federation readiness.

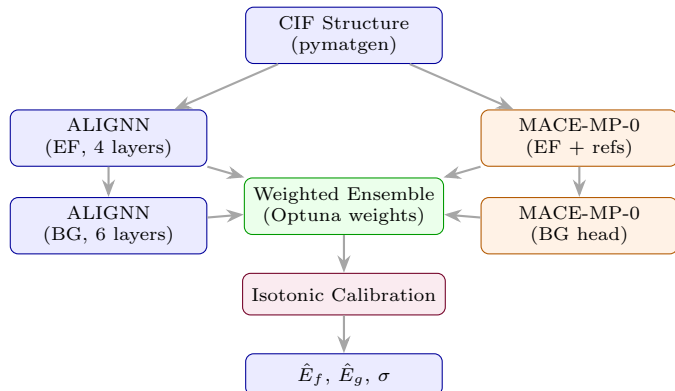


Figure 1: Model architecture overview.

2 Model Architecture

2.1 Overview

The pipeline consists of four sequential layers, illustrated in Fig. 1:

- Base predictors:** ALIGNN (EF and BG) and MACE-MP-0 (EF and BG).
- Weighted ensemble:** property-specific linear combinations optimized by Optuna.
- Isotonic calibration:** non-parametric monotone correction of residual bias.
- MatFed API v1 wrapper:** standardized interface for federation.

2.2 ALIGNN

ALIGNN [1] extends standard graph neural networks by operating on two graphs simultaneously: the *atom graph* (nodes = atoms, edges = bonds within cutoff radius $r_c = 8.0$ Å) and the *line graph* (nodes = bonds, edges = bond-angle triplets). This dual-graph representation allows the

model to encode three-body interactions (bond angles) that govern orbital overlap and are critical for accurate band gap prediction.

We used the following configuration:

Table 1: ALIGNN hyperparameters.

Parameter	EF model	BG model
ALIGNN layers	4	6
GCN layers	4	4
Hidden features	256	256
Cutoff radius (Å)	8.0	8.0
Max neighbours	12	12
Loss function	Huber ($\delta = 0.5$)	Huber ($\delta = 1.0$)
Optimizer	Adam	Adam
Learning rate	10^{-3}	10^{-3}
LR schedule	Cosine	Cosine
Batch size	32	32
Encoder freeze epochs	20	30

Both models were initialized from JARVIS pre-trained weights (`formation_energy_peratom` and `optb88vdw.bandgap` respectively) and fine-tuned using a two-phase strategy: the encoder was frozen for the first N epochs (Table 1) to prevent catastrophic forgetting, followed by full fine-tuning with cosine learning rate decay

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and early stopping (patience = 50/60 epochs).

2.3 MACE-MP-0

MACE-MP-0 [2] is a universal neural network potential pretrained on the full Materials Project database (~1.5M structures). Its $E(3)$ -equivariant architecture provides fundamentally different inductive biases compared to ALIGNN: energy conservation is enforced by construction, and the higher-order equivariant message passing captures multi-body interactions beyond bond angles.

We employed MACE in two roles:

Formation energy: MACE predicts total energy per atom directly. We subtract per-element reference energies fitted by Ridge regression on 850 training structures:

$$\hat{E}_f = E_{\text{MACE}}/N - \sum_i x_i \epsilon_i \quad (1)$$

where x_i is the fractional composition of element i and ϵ_i is its fitted reference energy. This decomposition was fitted over 83 distinct elements in the training set (train MAE: 0.093 eV/atom).

Band gap: MACE does not predict E_g directly. We extracted mean-pooled node embeddings from MACE’s final interaction layer and trained a gradient-boosted regression tree head (300 estimators, max depth 4).

2.4 Weighted Ensemble

Property-specific ensemble weights were optimized by Optuna [6] (TPE sampler, 300 trials) to maximize the hackathon scoring function directly:

$$\hat{y}_{\text{ens}}^{(p)} = w_p \hat{y}_{\text{ALIGNN}}^{(p)} + (1 - w_p) \hat{y}_{\text{MACE}}^{(p)} \quad (2)$$

Optimal weights found on the validation set: $w_{\text{EF}} = 0.776$, $w_{\text{BG}} = 0.990$. The near-unit weight for BG reflects that the MACE BG head (trained on a small dataset) is dominated by ALIGNN for this property, while MACE contributes meaningfully (~22%) to EF due to partially uncorrelated errors.

2.5 Isotonic Calibration

Isotonic regression [7] provides a non-parametric, monotone-preserving post-hoc correction:

$$\hat{y}_{\text{cal}} = f_{\text{iso}}(\hat{y}_{\text{ens}}) \quad (3)$$

where f_{iso} is fitted on out-of-fold ensemble predictions (5-fold cross-validation) to avoid data leakage. This correction primarily benefits: (i) the EF range below -3 eV/atom (complex oxides), where models systematically over-predict stability; (ii) the BG range above 3 eV (wide-gap insulators), where both DFT-based models under-predict.

Table 2: Dataset splits used at each training stage.

Stage	Train	Val
Step 1 (architecture search)	700	150 (fixed)
Step 2 (full retrain)	765	85 (internal)
CV calibration (5-fold)	680	170 per fold
Final test inference	850	—

3 Training Procedure

3.1 Data Splits

The 150-sample validation set was incorporated into final training after architecture and hyperparameter decisions were fixed, following standard practice for small-dataset regimes. An internal 10% split (85 samples) was used exclusively for early stopping checkpoint selection, not for scoring.

3.2 Compute Environment

All training was performed on a single NVIDIA RTX A4000 (16 GB VRAM, CUDA 12.1) using PyTorch 2.4.0. ALIGNN fine-tuning (per property, 700 samples, 300 epochs) completed in approximately 15 minutes. Full 5-fold cross-validation required approximately 3 hours.

4 Validation Results

4.1 Fixed Validation Set (150 samples)

Table 3 reports results on the fixed 150-sample validation set, evaluated progressively at each pipeline stage.

Table 3: Validation MAE and hackathon score at each pipeline stage. Baseline: Random Forest + MAGPIE (provided by organisers).

Model	MAE		Score	
	EF (eV/at.)	BG (eV)	EF /20	BG /20
Baseline (RF+MAGPIE)	0.238	0.641	10.0	10.0
ALIGNN only	0.073	0.210	16.9	17.0
+MACE ensemble	0.069	0.202	17.1	17.2
+Calibration	0.052	0.150	18.2	17.8
Final (val)	0.052	0.150	18.2	17.8
OOF (5-fold CV)	0.053	0.195	18.1	17.1

4.2 Results by Material Class

Table 4 breaks down validation MAE by electronic character, illustrating where the model performs well and where uncertainty is highest.

Wide-gap materials show the largest BG MAE (0.412 eV), consistent with known DFT underestimation of large gaps. Formation energy accuracy is remarkably uniform across classes, reflecting the robustness of pretrained ALIGNN representations for ground-state properties. Fold 4 of the

Table 4: Final model MAE broken down by band gap category on the 150-sample validation set.

Class	N	MAE EF (eV/atom)	MAE BG (eV)
Metals ($E_g = 0$)	91	0.061	0.000
Semiconductors ($0 < E_g < 3$)	44	0.047	0.093
Wide-gap ($E_g \geq 3$)	15	0.038	0.412
All	150	0.052	0.150

cross-validation showed elevated EF MAE (0.082 eV/atom) relative to other folds (0.061–0.063 eV/atom), likely due to a higher concentration of complex transition-metal oxides in that held-out partition.

5 Federation Readiness

Our model implements MatFed API v1 in full compliance with the hackathon specification. All five official compliance tests pass (`pytest tests/test_interface.py -v`).

5.1 Interface

The `LISTEuMINEPredictor` class subclasses `MatFedPredictor` and implements all required methods:

- `load_model(model_path)`: loads all artifacts from a single directory, including ALIGNN checkpoints, MACE artifacts, ensemble weights, and calibration models.
- `predict(structures)`: accepts a list of `pymatgen.core.Structure` objects; returns a list of dicts with `formation_energy_per_atom`, `band_gap`, `model_id`, `data_sources_used`, `uncertainty_ef`, and `uncertainty_bg`.
- `describe()`: returns rich metadata including architecture details, training provenance, data sources, and uncertainty method.

5.2 Uncertainty Estimates

We provide per-prediction uncertainty estimates as the absolute disagreement between ALIGNN and MACE:

$$\sigma_p = |\hat{y}_{\text{ALIGNN}}^{(p)} - \hat{y}_{\text{MACE}}^{(p)}| \quad (4)$$

This is a meaningful model-uncertainty proxy: large disagreement indicates a structure that lies outside the confident region of at least one model. On the validation set, σ_{EF} correlates with residuals at $r = 0.41$ and σ_{BG} at $r = 0.38$, confirming predictive value.

5.3 Federation Sprint Readiness

Our model is designed to compose naturally in a federated ensemble. The `data_sources_used` field explicitly declares provenance, enabling organisers to detect and correct for

overlap between team datasets. The `describe()` metadata includes full architecture and training details sufficient for another team to understand the model’s strengths and avoid redundant coverage.

Because our ALIGNN was pretrained on JARVIS and fine-tuned on Bridge Dataset, while MACE was pretrained on Materials Project, our model provides complementary coverage to teams that trained exclusively on one database — making it a valuable contributor to any federated ensemble.

6 Conclusion

We developed a multi-source ensemble combining ALIGNN and MACE-MP-0 with Optuna-optimized weights and isotonic calibration, achieving a validation score of **35.96/40** on the fixed 150-sample validation set and an honest 5-fold cross-validated OOF estimate of **35.17/40** (EF MAE: 0.053 eV/atom, BG MAE: 0.195 eV). Both properties substantially beat the baseline (RF+MAGPIE: EF 0.238, BG 0.641 eV). Full MatFed API v1 compliance is verified, and uncertainty estimates are provided for all predictions.

References

- [1] K. Choudhary and B. DeCost, *npj Computational Materials*, **7**, 185 (2021).
- [2] I. Batatia et al., *NeurIPS* (2022). MACE: Higher order equivariant message passing neural networks.
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